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Mechatronic Actuators Trainer

Fundamental Labs – Stepper Motor

V1.0 – August 1st, 2025

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please visit the web site at: http://www.quanser.com

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Caution

This equipment is designed to be used for educational and research purposes and is not intended for use by the public. The user is responsible for ensuring that the equipment will be used by technically qualified personnel only. Users are responsible for certifying any modifications or additions they make to the default configuration.

FCC Notice This device complies with Part 15 of the FCC rules. Operation is subject to the following two conditions: (1) this device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.
Note: This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment.

Industry Canada Notice This Class A digital apparatus complies with CAN ICES-3 (A). Cet appareil numérique de la classe A est conforme à la norme NMB-3 (A) du Canada.
Japan VCCI Notice This is a Class A product based on the standard of the Voluntary Control Council for Interference (VCCI). If this equipment is used in a domestic environment, radio interference may occur, in which case the user may be required to take corrective actions.

この装置は、クラス A 情報技術装置です。この装置を家庭環境で使用すると電波妨害を引き起こすことが あります。この場合には使用者が適切な対策を講ずるよう要求されることがあります。 VCCI–A

Waste Electrical and Electronic Equipment (WEEE)
This symbol indicates that waste products must be disposed of separately from municipal household waste, according to Directive 2012/19/EU of the European Parliament and the Council on waste electrical and electronic equipment (WEEE). All products at the end of their life cycle must be sent to a WEEE collection and recycling center. Proper WEEE disposal reduces the environmental impact and the risk to human health due to potentially hazardous substances used in such equipment. Your cooperation in proper WEEE disposal will contribute to the effective usage of natural resources.

电子信息产品污染控制管理办法 (中国 RoHS)

中国客户Quanser Consulting Inc. 关于关于限制在电子电气设备中使用某些有害成分的指令 (RoHS)。

CE Compliance CE

This product meets the essential requirements of applicable European Directives as follows:
- 2014/30/EU; Electromagnetic Compatibility Directive (EMC)

Warning: This is a Class A product. In a domestic environment this product may cause radio interference, in which case the user may be required to take adequate measures.

Stepper Motor Lab

Why Explore a Stepper Motor?

Stepper motors are a type of brushless dc motors that rotates in discrete steps. These steps can be achieved by sending electrical pulses to each of the coils. Because the motor moves in precise steps, position and speed can be tracked simply by counting the steps, no feedback sensors are required. Stepper motors can accurately be controlled to get to a specific speed or precisely hold a position while maintaining high torque. Because of how the phases are energized, steppers are also good at maintaining high torque at lower speeds, which can be challenging for other types of motors.

They are ideal for applications that require reliability and repeatability of movement and are easy to operate as an open loop control system. However, the torque of the motor decreases as the speed increase, making them better for lower speed operations.

Background

This lab is part of the Fundamentals labs for the Mechatronic Actuators Trainer. These labs will guide you through four different types of motors, so you can understand their operating principle and get you comfortable with using them. This will also include highlighting datasheets to learn how to interpret them and know what to look for.

Prior to starting this lab, please review the following concept review (located in Documents/Quanser/4_concept_reviews/),

- 4_concept_reviews/Mechatronics/**005_motor_stepper**.

Getting Started

In this lab, you will use the **Actuators Trainer** to learn about the basics of **stepper motors.** The lab will guide you through different ways of driving your stepper motor and the differences between them. You will experiment with them to see the differences in step sizes and speed. You will also be using the datasheet to validate your results.

Before you begin this lab, ensure that the following criteria are met:

- Make sure you have either the provided Stepper motor, or another bipolar stepper motor properly wired to be able to interface with the Actuators Trainer.
- Make sure the Actuators Trainer has been setup and tested. See the Quick Start Guide (Quanser/2_quick_start_guides/actuators_trainer) for details on this step.
- You are familiar with the basics of the programming language you are using for this lab.
- Have your development environment ready,
    - If using Python, we recommend using Visual Studio Code.